

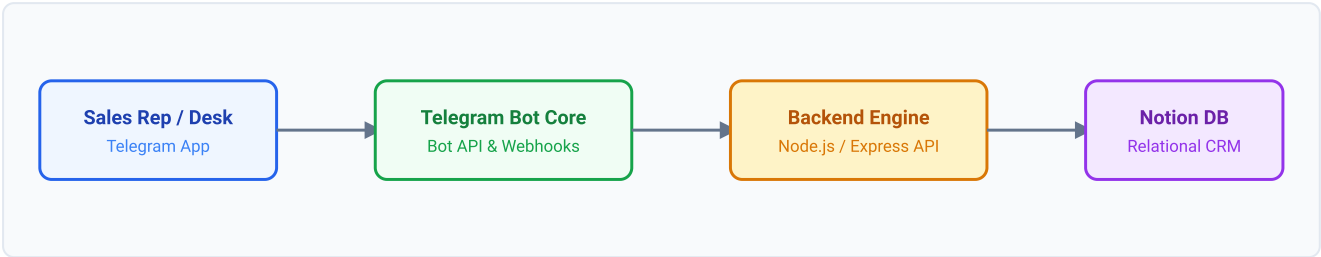
Telegram Web App & Notion CRM Integration

DETAILED TECHNICAL ARCHITECTURE & EXECUTION PLAN

TARGET AUDIENCE Sales Team & Execs	CORE TECH Node.js / Python + Notion API	DATABASE Notion Relational Workspace	CLIENT INTERFACE Telegram Mini App (TWA)
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1. System Architecture Overview

This project introduces a synchronized ecosystem linking field sales operations, walk-in front-desk management, and management oversight. The front-end operates as a lightweight, low-friction Telegram Web App (TWA) embedded directly within Telegram. The back-end service processes business logic, enforces duplicate prevention, and maintains real-time bi-directional synchronization with Notion, which functions as the executive CRM and data warehouse.



2. User Roles & Access Matrix

Role	Primary Interface	Key System Responsibilities	Access Control
Managers (3)	Notion Workspace	Pipeline oversight, performance tracking, conflict resolution, global sales analytics.	Full Workspace Admin Access
Sales Reps (6)	Telegram Mini App	TIN verification, lead registration, follow-up logging, deposit proof upload.	Restricted to self-assigned records via Bot
Front Desk (1)	Telegram Web App	Register walk-in customers, trigger rotational queue assignment to active reps.	Specialized Receptionist Interface

3. Notion Database Schema & Architecture

To enable drill-down reports for managers, the Notion workspace is structured into six linked databases. This normalized architecture ensures data integrity and high-performance querying via the Notion API.

Database Name	Key Fields / Properties	Relations & Formulae
Sales Reps	Telegram ID (Unique ID), Full Name, Phone, Status (Active/Inactive), Total Sales Count	Relation to <i>Accounts</i> , <i>Deals</i> , and <i>Sales Logs</i> .
Accounts (Customers)	Company Name, TIN Number (Unique Index), Address, Industry, Assigned Date	Relation to <i>Sales Reps</i> (Owner), <i>Contacts</i> , and <i>Deals</i> .
Deals / Pipeline	Deal Title, Stage (New, Contacted, Proposal, Won, Lost), Amount, Deposit Ref #	Relation to <i>Accounts</i> , <i>Sales Reps</i> , and <i>Payment Proofs</i> .
Contacts	Contact Name, Phone, Email, Position, Primary Contact Flag	Relation to <i>Accounts</i> .
Sales Logs	Interaction Type (Call, Meeting, Note), Activity Date, Note Content, Location	Relation to <i>Deals</i> and <i>Sales Reps</i> .
Rotational Queue	Rep ID, Order Position, Availability Status, Last Assigned Timestamp	Used by Front Desk algorithm for walk-in routing.

4. Core Technical Workflows

Workflow A: Sales Rep Onboarding - Rep launches Telegram Bot using <code>/start</code>. - Bot queries Notion Sales Reps DB via Telegram User ID. - If ID is missing, Bot initiates setup form (Name, Phone, Role). - Backend creates record in Sales Reps DB and sets status to "Pending Approval". - Manager receives alert in Notion/Telegram to approve access.	Workflow B: TIN Conflict Verification - Rep initiates "New Lead" in Telegram Web App. - App prompts for Tax Identification Number (TIN). - Backend queries Notion Accounts DB matching TIN. Match Found: Returns owner name and blocks registration. No Match: Unlocks registration form and reserves TIN.
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Conflict Prevention Protocol: If a sales rep enters an existing TIN, the system immediately returns: *"Company registered under [Sales Rep Name] on [Date]. Duplicate registration blocked. Please contact [Sales Rep Name] for internal transfers."*

5. Specialized Workflows

Front Desk Walk-In Assignment Automated round-robin assignment system for incoming leads. • Receptionist inputs Walk-in Company Details and TIN into Front Desk Web App view. • System checks TIN for existing owner. If clear, queries Rotational Queue DB for next available rep. • Assigns lead to designated rep, logs transaction, and advances rep queue position. • Instant push notification sent to Rep's Telegram: <i>"New Walk-In assigned!"</i>	Purchase & Payment Proof Logging Post-sale verification and accounting handover. • Sales Rep selects deal in Mini App and clicks "Log Purchase". • Rep inputs Deposit Reference Number and attaches receipt/bank transfer image. • Image is uploaded to cloud storage (S3) and attached to Notion Deals DB. • Stage updates to "Payment Pending Verification" and alerts Manager.
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6. Manager Workspace & Navigation Architecture

Notion serves as the executive cockpit. The workspace is built around dynamic relational views, allowing managers to seamlessly inspect rep activities without switching contexts.

Dashboard View	Layout Components	Executive Purpose
Sales Rep Directory	Gallery View with Rollup properties showing active deal count, monthly revenue, last activity timestamp.	1-Click drill-down into an individual sales rep's full activity log and accounts.
Master Pipeline Board	Kanban Board grouped by Stage, filtered by Rep, Date, or Deal Value.	Visual tracking of company-wide sales progress and stage velocity.
Audit & Conflict Log	Table View showing failed TIN attempts, reassignments, and deposit approvals.	Ensures organizational policy compliance and resolves disputes.

7. Automated Notification Engine

A background worker (Cron / Node-Cron) periodically queries Notion databases to trigger automated Telegram notifications to keep sales reps engaged and proactive.

- **Follow-Up Alerts:** Daily 9:00 AM push notification detailing leads scheduled for follow-up today.
- **Inactivity Warning:** Triggers if an active deal has no logged activity in 7 days.
- **Stage Movement Updates:** Manager alerts when a deal transitions to "Payment Proof Submitted".

8. Implementation & Deployment Roadmap

Phase	Milestones	Deliverables	Timeline
Phase 1	Notion Database Architecture	Setup databases, relations, rollups, formula fields, and access permissions.	Week 1
Phase 2	Backend & API Integration	Node.js/Python server, Notion API service layer, Telegram Bot authentication.	Weeks 2–3
Phase 3	Telegram Mini App (TWA)	Frontend interface for TIN lookup, lead creation, follow-ups, and image uploads.	Weeks 3–4
Phase 4	Rotational Queue & Triggers	Front Desk interface, round-robin algorithm, automated reminder notifications.	Week 5
Phase 5	Testing & Deployment	UAT with 6 reps + 1 receptionist, performance optimization, user training.	Week 6

9. Security, Authentication & Data Integrity

Since Telegram handles client access, security relies on Telegram WebApp initData cryptographic verification. Every HTTP request made from the WebApp passes a signature derived from the Bot Token, ensuring users cannot forge another sales rep's Telegram ID.

Critical Security Enforcement: All write operations (creating leads, modifying deals) require token validation at the backend server. The backend directly checks Notion permissions prior to performing any mutating database queries.

10. Next Steps for Execution

- Notion Workspace Preparation:** Create an integration token under `notion.so/my-integrations` and grant access to the root CRM page.
- Bot Father Provisioning:** Register the Telegram Bot via @BotFather and configure the Mini App domain endpoint.
- Environment Setup:** Provision backend hosting (e.g., Render, Railway, or AWS) with Node.js / Python environment.